

Tribhuvan University
Institute of Science and Technology
2073

Bachelor Level / Third Year /Six Semester/Science
Computer Science and Information Technology-(CSc.363)
(Image Processing)

Full Marks: 60
Pass Marks: 24
Time: 3 hours.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempt any ten questions.

1. Differentiate between digital image representation and digital image processing.
2. Explain the fast fourier transform with example.
3. What do you mean by image enhancement? Explain the Histogram Modeling with example.
4. What do you mean by image coding? Explain the process of predictive coding.
5. Explain the features extraction with appropriate example. Why feature extraction is important in image processing? Explain.
6. Explain the edge detection using templates with practical example.
7. Explain the segmentation by thresholding. Explain how can you apply segmentation in line detection.
8. Explain the Hadamard transform with example.
9. Explain the predictive and inter-frame coding with example.
10. Differentiate between Hopfield nets and hamming nets.
11. Differentiate between:
 - a) Sampling and quantization
 - b) High pass and band pass filtering